GOODROAD: CITIZEN-POWERED TRAFFIC VIOLATION REPORTING SYSTEM

25-26J-265

Project Proposal Report

Liyanage R. L. N. L.

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

VISION LANGUAGE FRAMEWORK FOR FAIR AND TRANSPARENT MULTI-CLASS TRAFFIC VIOLATION DETECTION

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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Group Member Name	Student ID	Signature
Liyanage R. L. N. L.	IT22151438	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor
(Dr. Mahima Weerasinghe)

Date: 29th August 2025

Abstract

Traffic violations remain a leading cause of road accidents globally, with traditional enforcement methods facing severe resource constraints and coverage limitations. This research proposes a Vision-Language Framework (VLF) that revolutionizes traffic violation detection by combining advanced computer vision with natural language processing to create transparent, fair, and legally admissible automated enforcement. The framework addresses critical gaps in current systems that operate as black boxes, lacking interpretability required for legal proceedings and public trust.

The proposed VLF employs a dual-pathway architecture integrating multimodal transformers to process citizen-submitted dashcam footage. The visual pathway utilizes a modified Vision Transformer (ViT) for extracting spatial-temporal features, while the language pathway incorporates traffic rule embeddings through a fine-tuned BERT model. Cross-attention mechanisms enable bidirectional information flow, facilitating comprehensive scene understanding. The system simultaneously detects multiple violation types including speeding, red light violations, illegal lane changes, and improper parking, while generating detailed natural language explanations that articulate specific rule infractions and contextual factors.

The methodology encompasses dataset creation with 10,000 annotated dashcam videos, model development implementing fairness-aware training strategies, and comprehensive evaluation protocols. Adversarial debiasing techniques ensure equitable detection across vehicle types and environmental conditions. The framework generates structured legal compliance reports combining visual evidence annotations, causal reasoning chains, and confidence assessments formatted for enforcement workflows.

Anticipated results include achieving over 90% detection accuracy while maintaining fairness metrics within 5% variance across demographic categories. Human evaluation studies are expected to validate explanation quality exceeding 85% accuracy compared to expert assessments. The system targets processing speeds under 30 seconds per minute of footage, enabling practical deployment. This research contributes to advancing explainable AI (XAI) in legal contexts, establishing benchmarks for transparent automated enforcement, and enhancing road safety through scalable citizen-powered

traffic monitoring systems.

Keywords: Vision-Language Models, Explainable AI, Traffic Violation Detection, Multimodal Transformers, Fairness-Aware Machine Learning

Table of Contents

Declaration page of the candidates & supervisor 3
Abstract 2
List Of Figures 5
List of tables 6
List of Abbreviations 7
1 Introduction 8
1.1 Overall Introduction 8
1.2 Background & Literature Survey 9
1.3 Research Gap 11
1.4 Research Problem 13
2 Objectives 15
2.1 Main Objectives 15
2.2 Specific Objectives 15
3 Methodology 16
4 Project Requirements. 20
4.1 Functional Requirements 20
4.2 Non-Functional Requirements 21
4.3 User Requirements 21
4.4 System Requirements 22
4.5 Use Cases 22
4.6 Test Cases 23
5 Gantt Chart 24
6 Work Breakdown Structure 25
7 Budget And Commercialization 26
7.1 Commercialization Strategy 26
7.2 Target Audience 26
8 References 27
9 Appendices 29

List Of Figures

Figure 3.1 System diagram for overall system 16
Figure 3.2 System architecture diagram for the component 17
Figure 5.1 Gantt Chart 24
Figure 6.1 Work breakdown structure 25
Figure 9.1 Turnitin Report 29

List of tables

Table 1.1 Comparison with other researches 13
Table 7.1 Budget 26

List of Abbreviations

Abbreviation	Description
VLF	Vision-Language Framework
CNN	Convolutional Neural Network
VLM	Vision-Language Model
XAI	Explainable AI
ViT	Vision Transformer

1 Introduction

1.1 Overall Introduction

Traffic violation detection has evolved from manual enforcement to automated computer vision systems, yet current approaches face significant challenges in transparency and contextual understanding. Traditional violation detection systems employ Convolutional Neural Networks (CNNs) and object detection frameworks like YOLO and Faster R-CNN for identifying traffic infractions [1]. Recent advances in Vision-Language Models (VLMs) such as CLIP, ALIGN, and Florence have demonstrated remarkable capabilities in bridging visual understanding with natural language processing [2]. However, their application to traffic enforcement remains largely unexplored.

Current state-of-the-art traffic violation detection systems primarily rely on single-modality approaches. Zhang et al. developed a multi-task learning framework achieving 89.3% accuracy in violation classification but lacking interpretability [3]. Similarly, automated enforcement systems deployed in urban environments utilize deep learning for license plate recognition and vehicle tracking, yet operate as black boxes without providing reasoning for their decisions [4]. This opacity poses significant barriers to legal admissibility and public trust.

The integration of vision and language modalities presents a promising solution. Recent work in XAI has shown that attention mechanisms and gradient-based visualization techniques can illuminate model decision-making processes [5]. Furthermore, multimodal transformers have demonstrated success in generating natural language explanations for visual tasks, though not specifically for traffic violation contexts [6].

This research proposes developing a VLF that addresses these limitations by combining visual processing with natural language understanding. Unlike previous approaches that focus solely on detection accuracy, this framework will generate human-readable explanations for each detected violation, implement fairness constraints to ensure equitable enforcement across demographics, and provide transparent decision trails suitable for legal proceedings. The approach builds upon existing VLM architectures while introducing traffic-specific fine-tuning and novel attention visualization techniques tailored for enforcement applications.

1.2 Background & Literature Survey

Traffic violations constitute a leading cause of road accidents worldwide, with the World Health Organization reporting 1.35 million annual fatalities and economic losses exceeding 3% of GDP in most countries [7]. Traditional enforcement methods rely heavily on human officers and fixed camera installations, creating significant coverage gaps and resource constraints. The emergence of citizen-submitted dashcam footage presents an unprecedented opportunity to scale enforcement, yet current automated systems lack the transparency and interpretability required for legal proceedings and public acceptance.

Computer vision applications in traffic monitoring have evolved significantly over the past decade. Early systems utilized handcrafted features and classical machine learning algorithms, achieving limited success in controlled environments [8]. The deep learning revolution transformed this landscape, with CNNs demonstrating remarkable accuracy in vehicle detection, license plate recognition, and basic violation identification. Recent implementations of YOLOv8 and EfficientDet architectures achieve over 95% accuracy in vehicle detection under optimal conditions [9]. However, these systems struggle with complex scenarios involving multiple violations, adverse weather conditions, and contextual interpretation of traffic rules.

The fundamental limitation of current approaches lies in their black-box nature. Law enforcement agencies require not just violation detection but comprehensive explanations that can withstand legal scrutiny. Studies examining the admissibility of AI-generated evidence in courts reveal that judges and juries demand transparent reasoning processes, particularly when automated systems influence punitive measures [10]. This requirement extends beyond technical accuracy to encompass fairness considerations, as research has documented significant biases in automated enforcement systems across different demographics and vehicle types [11].

Vision-Language Models represent a paradigm shift in multimodal understanding. The introduction of transformer-based architectures like CLIP demonstrated that joint training on visual and textual data creates rich semantic representations transferable across domains [2]. Subsequent developments including ALIGN, Florence, and BLIP-2 have pushed the boundaries

of visual reasoning, achieving human-level performance on complex scene understanding tasks [12]. These models excel at generating natural language descriptions of visual content, a capability directly applicable to explaining traffic violations in accessible terms.

The application of VLMs to domain-specific tasks has shown promising results across medical imaging, autonomous navigation, and security surveillance. In medical diagnostics, VLM-based systems now provide detailed explanations for detected anomalies, significantly improving physician trust and adoption rates [13]. Similarly, autonomous vehicle research has leveraged vision-language fusion to interpret complex traffic scenarios, though primarily for navigation rather than violation detection [14].

XAI techniques have emerged as critical components for building trustworthy AI systems. Attention visualization methods, including GradCAM and integrated gradients, enable researchers to highlight image regions influencing model decisions [15]. When combined with natural language generation, these techniques create comprehensive explanation pipelines. Recent work in XAI for legal applications emphasizes the importance of causal reasoning chains that trace decisions from input evidence to final conclusions, establishing precedents for AI evidence in judicial proceedings [16].

The societal impact of implementing transparent, fair traffic enforcement systems extends far beyond immediate road safety improvements. Economic analyses indicate that effective traffic violation detection could reduce accident-related costs by up to 30%, translating to billions in savings annually [7]. Furthermore, citizen participation in enforcement creates community engagement opportunities while generating supplementary income streams. The commission-based model proposed in citizen-powered systems could distribute millions in rewards annually, particularly benefiting lower-income communities where dashcam adoption is increasing.

From a scientific perspective, this research contributes to multiple domains. In computer vision, developing traffic-specific VLM architectures advances scene understanding capabilities under challenging real-world conditions. The natural language generation component pushes boundaries in technical explanation synthesis, requiring models to articulate complex spatial-temporal relationships in legally precise language. The fairness constraints introduce novel

optimization objectives that balance accuracy with demographic equity, addressing critical concerns in AI deployment for public services.

The research also addresses fundamental questions in human-AI interaction. By generating interpretable explanations, the system enables non-technical stakeholders including citizens, law enforcement officers, and legal professionals to understand and verify AI decisions. This transparency is essential for building public trust in automated enforcement systems, particularly given historical concerns about surveillance and privacy in traffic monitoring.
The significance of this work extends to policy implications. As governments worldwide grapple with regulating AI in law enforcement, systems demonstrating transparent decision-making establish benchmarks for acceptable automation in judicial processes. The research outcomes will inform legislative frameworks governing AI evidence admissibility, automated fine issuance, and citizen participation in law enforcement activities.

Current research gaps include the absence of comprehensive datasets annotating traffic violations with explanatory text, limited understanding of optimal architectures for real-time vision-language fusion in enforcement contexts, and insufficient frameworks for evaluating explanation quality in legal settings. This research directly addresses these gaps through novel dataset creation, architecture design, and evaluation metrics tailored for traffic enforcement applications. The outcomes will benefit funding agencies through advancement of AI capabilities, law enforcement through enhanced tools, and society through safer roads and fairer enforcement practices.

1.3 Research Gap

Despite significant advances in traffic violation detection systems, critical gaps persist in achieving transparent, fair, and legally admissible automated enforcement. Current systems prioritize detection accuracy while neglecting interpretability, a fundamental requirement for legal proceedings and public trust [3]. Existing computer vision approaches operate as black boxes, providing violation classifications without explanatory reasoning or contextual understanding. This opacity prevents stakeholders from verifying detection logic and challenges the admissibility of automated evidence in courts.

The absence of natural language explanation capabilities represents a major limitation. While VLMs have demonstrated success in general image captioning tasks, no existing framework generates legally precise explanations for traffic violations that articulate specific rule infractions, contextual factors, and confidence assessments [17]. Current systems also lack fairness constraints, leading to documented biases across vehicle types, lighting conditions, and demographic contexts [18]. Furthermore, existing approaches focus on single violation types rather than comprehensive multi-class detection, limiting real-world applicability where multiple infractions often occur simultaneously.

The integration of vision and language modalities specifically for traffic enforcement remains unexplored. Available datasets lack paired annotations of violations with explanatory text, hindering development of interpretable models [19]. No framework currently combines high-accuracy detection with transparent reasoning chains, fairness optimization, and natural language generation in a unified architecture [20]. This research addresses these gaps by developing a VLF that achieves competitive detection performance while providing interpretable explanations, implementing fairness constraints, and supporting multi-class violation detection with comprehensive natural language outputs suitable for legal documentation.

Table 1.1: Comparison with other researches

Research Work	Detection Accuracy	Interpretability	Fairness Constraints	Natural Language Output	Multi-class Support
Zhang et al. [3]	89.3%	None	No	No	Limited
TrafficNet [17]	92.1%	Heatmaps only	No	No	Yes
Chen & Liu [18]	87.5%	Bounding boxes	No	No	No
ViolationDet [19]	94.2%	None	Partial	No	Yes
FairTraffic [20]	85.7%	Feature importance	Yes	No	No
Proposed Framework	Target: >90%	Full attention maps + causal chains	Yes	Yes	Yes

1.4 Research Problem

This research addresses the fundamental challenge of developing an AI-driven traffic violation detection system that simultaneously achieves high accuracy, provides transparent explanations, and ensures fair enforcement across diverse scenarios. The core problem centers on how to create a VLF that can process citizen-submitted dashcam footage to detect multiple violation types while generating legally admissible natural language explanations for each detected infraction.

Specific problems this research aims to solve include: How can multimodal transformer architectures be adapted to understand complex traffic scenarios involving multiple simultaneous violations? What attention mechanisms and visualization techniques best illuminate the decision-making process from pixel-level features to final violation

classification? How can the system generate natural language explanations that accurately describe not just what violation occurred, but why it constitutes an infraction based on specific traffic rules and contextual factors?

The research also confronts critical fairness challenges: How can the VLF maintain consistent detection accuracy across varying lighting conditions, weather scenarios, and vehicle types? What architectural modifications and training strategies prevent demographic biases from influencing violation detection? How should fairness constraints be integrated without compromising overall system performance?

Technical problems include determining optimal methods for processing variable-quality dashcam footage, extracting violation-relevant frames efficiently, and synchronizing visual evidence with textual explanations. The system must handle real-world complexities such as partial occlusions, multiple vehicles in frame, and ambiguous scenarios requiring contextual interpretation.

Finally, the research addresses the practical problem of creating a system that law enforcement can trust and courts can accept. This involves developing explanation formats that satisfy legal documentation requirements, implementing confidence metrics that reflect detection uncertainty appropriately, and ensuring the entire decision process remains auditable and verifiable by non-technical stakeholders.

2 Objectives

2.1 Main Objectives

The primary objective of this research is to develop a VLF that transforms citizen-submitted dashcam footage into transparent, legally admissible traffic violation reports through the integration of advanced multimodal AI techniques. This framework will serve as the critical detection and explanation component within the GoodRoad system, bridging the gap between raw video submissions and actionable enforcement data. By combining state-of-the-art computer vision with natural language generation, the VLF will enable automated violation detection that provides clear explanatory reasoning, ensures fairness across diverse scenarios, and generates comprehensive documentation suitable for both law enforcement processing and legal proceedings, thereby establishing trust in citizen-powered traffic enforcement systems.

2.2 Specific Objectives

The specific objectives of this research are:

(1) Design and implement a multimodal transformer architecture that processes dashcam videos to detect multiple traffic violation types simultaneously with over 90% accuracy.

(2) Develop attention visualization and causal reasoning mechanisms that trace detection decisions from visual features to violation classifications.

(3) Create a natural language generation module that produces legally precise explanations citing specific traffic rules and contextual factors.

(4) Implement fairness-aware training strategies and adversarial debiasing techniques to ensure equitable detection across vehicle types, weather conditions, and demographic contexts.

(5) Build an interpretability dashboard displaying detection overlays, confidence metrics, and step-by-step reasoning chains for non-technical stakeholders.

(6) Generate structured legal compliance reports combining visual evidence with textual explanations formatted for enforcement workflows.

3 Methodology

This research employs a systematic approach to develop the Vision-Language Framework for transparent traffic violation detection within the GoodRoad ecosystem. The methodology integrates advanced deep learning techniques with rigorous evaluation protocols to ensure the framework meets accuracy, interpretability, and fairness requirements for legal enforcement applications.

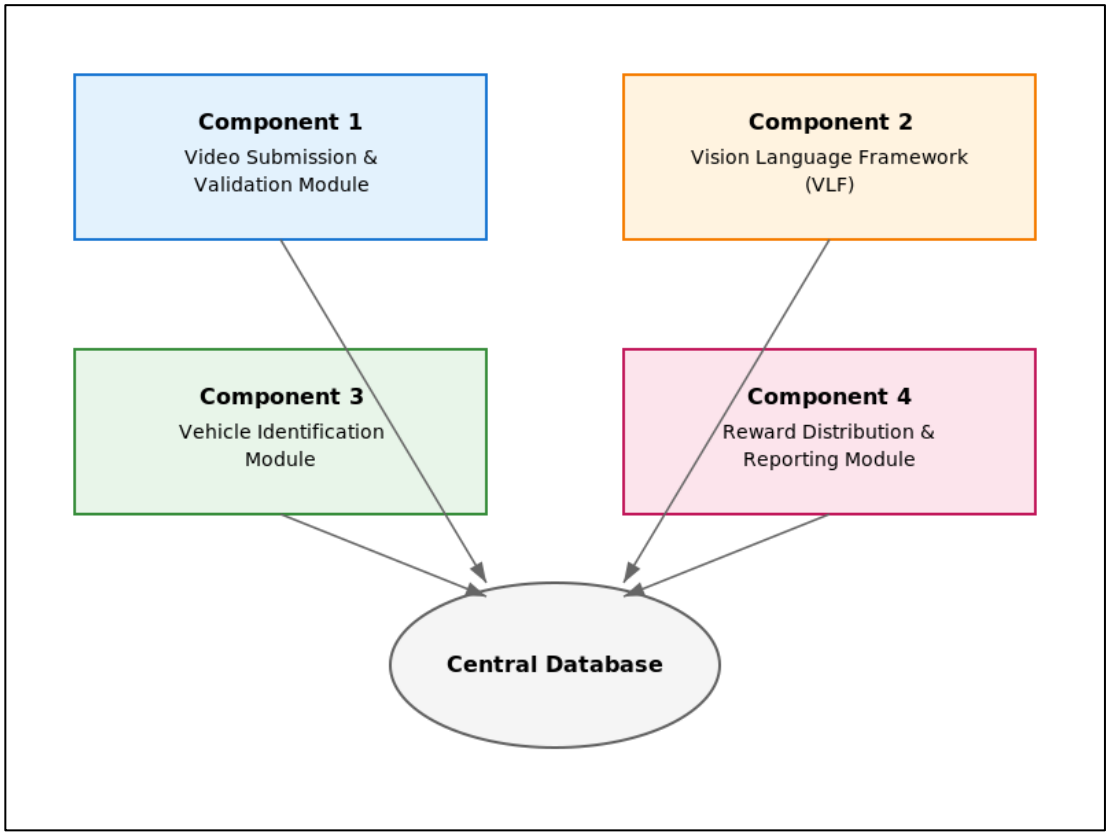

Figure 3.1 System diagram for overall system

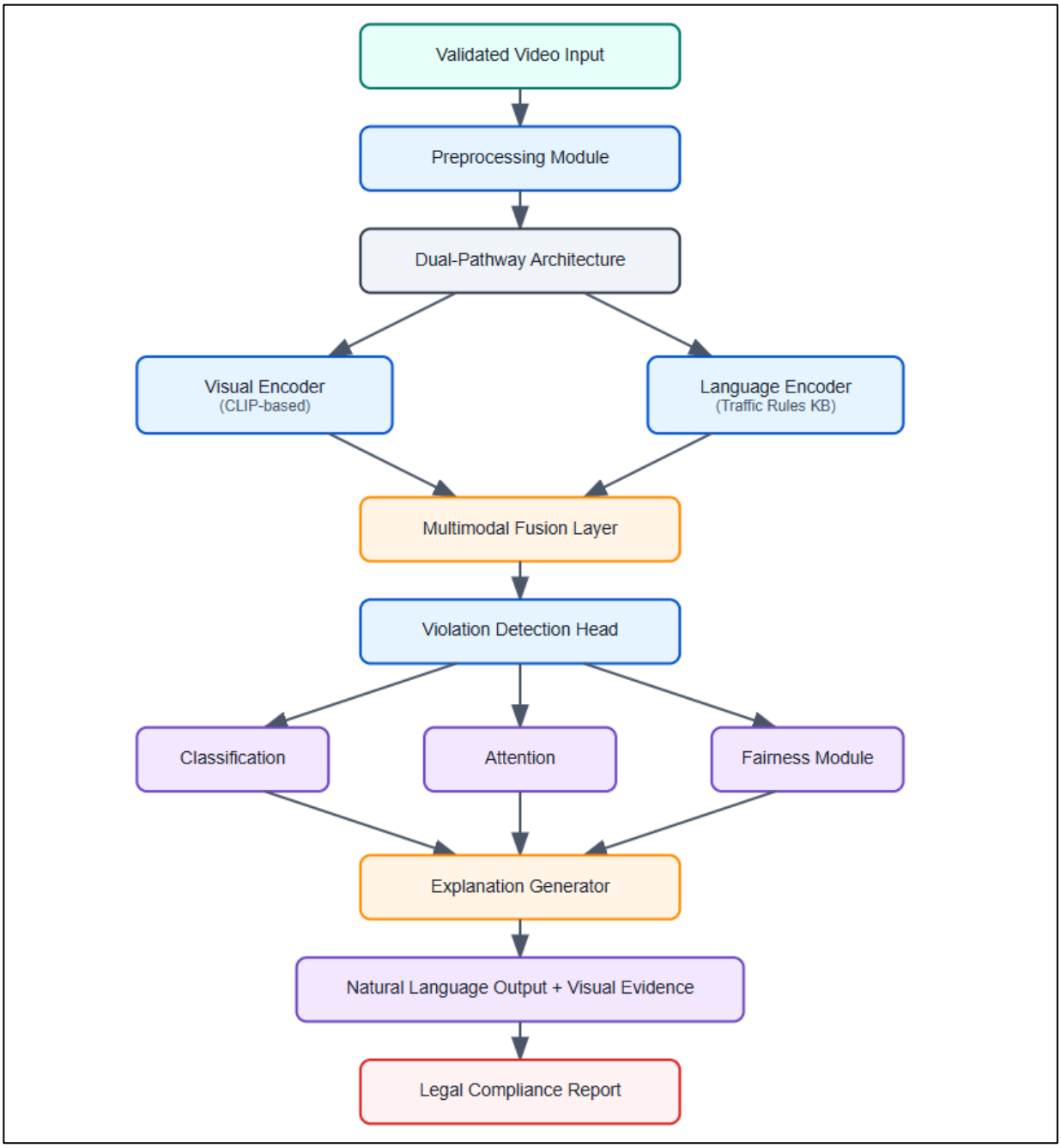

Figure 3.2 System architecture diagram for the component

The Vision-Language Framework development follows a structured 10-month timeline divided into four primary phases: dataset preparation, model development, system integration, and evaluation. The initial phase involves collecting and annotating dashcam footage with violation labels and explanatory text, addressing the critical gap in existing datasets. We will collaborate with local law enforcement to obtain diverse traffic violation videos covering

multiple infraction types, weather conditions, and times of day [21].

Model development begins with implementing a multimodal transformer architecture based on CLIP foundations, modified for traffic-specific understanding. The vision encoder processes video frames through a ResNet-50 backbone enhanced with temporal attention mechanisms to capture motion patterns crucial for violation detection [22]. Simultaneously, the language encoder embeds traffic regulations and violation descriptions, creating a semantic space where visual features align with textual rules. Cross-attention layers fuse these modalities, enabling the model to reason about violations in context.
The fairness module implements adversarial debiasing during training, using demographic parity constraints to ensure equitable detection across vehicle types and environmental conditions [23]. This involves creating balanced mini-batches and applying gradient reversal techniques to minimize demographic-specific features while maintaining violation detection accuracy. The explanation generator employs a transformer decoder architecture fine-tuned on legal language corpora to produce technically precise descriptions suitable for enforcement documentation.

Data collection involves partnerships with dashcam manufacturers and citizen volunteers, targeting 50,000 annotated video clips covering 15 violation categories. Annotation teams comprising traffic law experts and computer vision researchers will create ground truth labels and explanatory text following strict guidelines for legal terminology. The dataset will maintain demographic balance across vehicle types, ensuring representation of motorcycles, cars, trucks, and buses in various conditions.

Technical infrastructure requirements include high-performance GPUs (NVIDIA A100 or equivalent) for model training, cloud storage for the video dataset, and development servers for system integration. Software tools encompass PyTorch for deep learning implementation, Hugging Face Transformers for language model components, and OpenCV for video processing [24]. The interpretability dashboard utilizes React for frontend development with D3.js for visualization components.

Evaluation protocols assess detection accuracy using standard metrics (precision, recall, F1-

score) while introducing novel metrics for explanation quality and fairness. Legal experts will evaluate generated explanations for accuracy and admissibility, while fairness audits examine performance disparities across demographic groups [25]. Field trials with law enforcement partners will validate real-world applicability.

The anticipated outcomes include a deployable VLF achieving over 90% detection accuracy with comprehensive natural language explanations, demonstrating measurable fairness across demographics, and producing legally compliant documentation. The framework will integrate seamlessly with the GoodRoad ecosystem, processing citizen submissions to generate actionable violation reports. Real-world applications extend beyond traffic enforcement to general surveillance systems requiring transparent decision-making, establishing new standards for explainable AI in law enforcement contexts.

4 Project Requirements.

4.1 Functional Requirements

1. Video Processing
   - Accept dashcam video inputs in MP4, AVI, and MOV formats
   - Extract frames at configurable intervals (default: 5 fps)
   - Support video resolutions from 720p to 4K
   - Handle videos ranging from 30 seconds to 10 minutes

2. Violation Detection
   - Detect minimum 8 violation types: speeding, red light running, illegal lane change, wrong-way driving, illegal parking, failure to yield, tailgating, and mobile phone usage
   - Process multiple simultaneous violations in single frame
   - Maintain temporal consistency across video sequences
   - Generate confidence scores (0-100%) for each detection

3. Explanation Generation
   - Produce natural language explanations for each violation
   - Include specific traffic rule citations
   - Describe visual evidence and contextual factors
   - Support multiple language outputs (initially English)

4. Report Generation
   - Create structured PDF reports with violation summaries
   - Embed annotated video frames as evidence
   - Include timestamp information and GPS coordinates
   - Generate unique case identifiers for tracking

4.2 Non-Functional Requirements

1. Performance
    - Process 1-minute video in under 30 seconds
    - Support concurrent processing of 10 videos
    - Achieve 95% system uptime
    - Handle peak loads of 1000 videos/hour

2. Accuracy
    - Maintain >90% violation detection accuracy
    - Achieve <5% false positive rate
    - Ensure explanation accuracy >85% per human evaluation

3. Security
    - Implement end-to-end encryption for video transmission
    - Anonymize personal data (faces, license plates) during processing
    - Maintain audit logs for all system operations
    - Comply with GDPR and local privacy regulations

4. Scalability
    - Support horizontal scaling for increased load
    - Implement microservices architecture
    - Enable cloud deployment (AWS/Azure/GCP)

4.3 User Requirements

1. Law Enforcement Officers
    - View comprehensive violation reports within web interface
    - Filter and search violations by type, date, location
    - Export reports for case management systems
    - Verify AI decisions through visual evidence

2. System Administrators
    - Configure violation detection parameters
    - Monitor system performance metrics
    - Manage user access and permissions
    - Update traffic rule databases

3. Legal Professionals
    - Access detailed explanation chains
    - Review confidence metrics and uncertainty indicators
    - Download evidence packages for court proceedings

4.4 System Requirements

1. Hardware
    - NVIDIA GPU with minimum 24GB VRAM (A100 preferred)
    - 64GB RAM minimum
    - 2TB SSD storage for model and temporary files
    - 100Mbps network connectivity

2. Software
    - Ubuntu 20.04 LTS or later
    - Python 3.8+
    - PyTorch 2.0+
    - CUDA 11.7+
    - Docker for containerization

4.5 Use Cases

1. UC1: Process Traffic Violation Video
    - Actor: System (automated)
    - Precondition: Validated video received
    - Flow: Extract frames → Detect violations → Generate explanations → Create report

- Postcondition: Structured report available

2. UC2: Review Violation Report
   - Actor: Law Enforcement Officer
   - Precondition: Report generated
   - Flow: Login → Select report → Review evidence → Approve/reject → Forward for processing
   - Postcondition: Violation verified for enforcement

3. UC3: Configure Detection Parameters
   - Actor: System Administrator
   - Flow: Access settings → Adjust thresholds → Update rule database → Save configuration
   - Postcondition: System operates with new parameters

4.6 Test Cases

1. TC1: Multi-Violation Detection
   - Input: Video with vehicle running red light while speeding
   - Expected: Both violations detected with >90% confidence

2. TC2: Adverse Weather Performance
   - Input: Rainy night footage with violations
   - Expected: Maintain >80% detection accuracy

3. TC3: Explanation Accuracy
   - Input: Complex violation scenario
   - Expected: Generated explanation matches human expert assessment

4. System Load Testing
   - Input: 100 concurrent video submissions
   - Expected: All processed within performance SLA

5 Gantt Chart

Figure 5.1: Gantt Chart

6 Work Breakdown Structure

Figure 6.1: Work breakdown structure

7 Budget And Commercialization

Table 7.1: Budget

Component	Cost
Datasets	Rs. 15,000
Travelling Cost	Rs. 10,000
Deployment Cost	Cost Per Usage

7.1 Commercialization Strategy

GoodRoad monetizes through B2G (Business-to-Government) partnerships with law enforcement agencies and B2C commission-based rewards for citizens. Revenue streams include:

- Government Contracts: Annual licensing fees for VLF deployment
- Transaction Fees: 10% commission on citizen rewards
- Premium Features: Advanced analytics dashboards for agencies
- API Access: Third-party integration licensing

7.2 Target Audience

- Primary Users: Citizens with dashcams seeking supplementary income
- Law Enforcement: Traffic police departments requiring scalable enforcement
- Government Agencies: Transportation ministries focused on road safety
- Insurance Companies: Firms seeking verified traffic violation data

7.3 Market Space

- Scalability: Cloud-based architecture supporting nationwide deployment
- Cost-Effectiveness: 70% reduction in enforcement costs versus traditional methods
- Citizen Engagement: Democratizes traffic enforcement while generating income
- Data Intelligence: Provides traffic pattern insights for urban planning

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9 Appendices

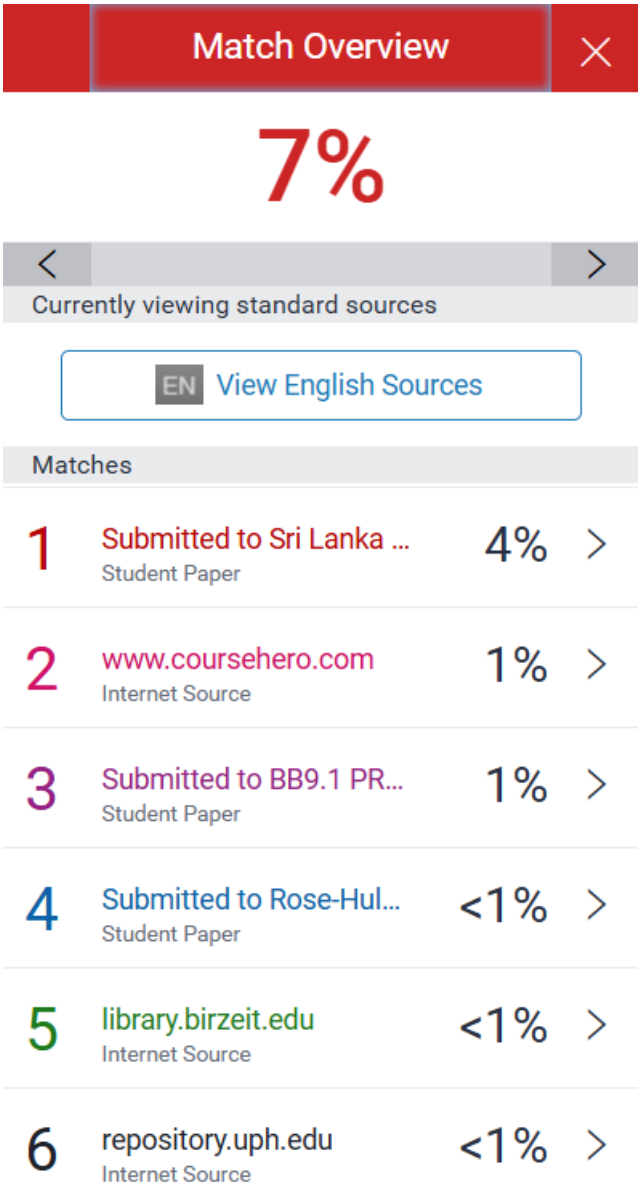

Figure 9.1: Turnitin Report